

Efficient quantum simulation of random-coupling models by spectral amplification

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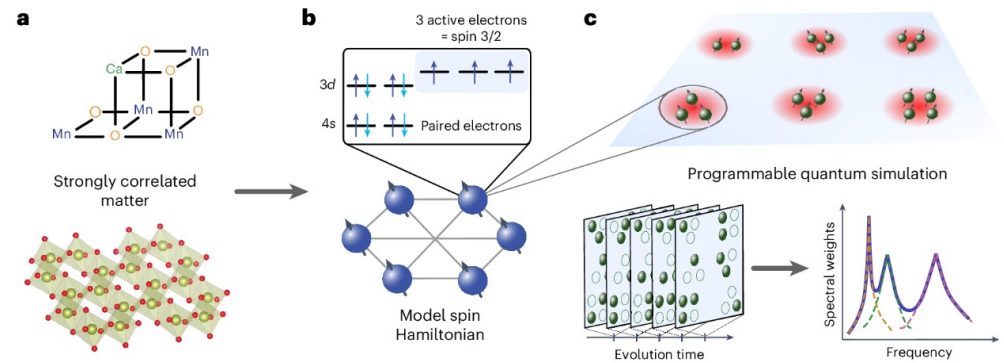
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Quantum simulation

$$|\psi(t)\rangle = e^{-iHt} |\psi(0)\rangle$$

- many-body dynamics
- energy and spectral estimation
- quantum chemistry
- material science



N. Maskara et al., Nature Physics 2025 Fig.1

No fast-forwarding theorem

- Lower bound on query complexity for quantum signal processing^[1]

$\mathcal{O}(d\|H\|_{\max} t + \log(1/\epsilon)) \in \Omega(t)$ queries to the d -sparse Hamiltonian oracle for error ϵ

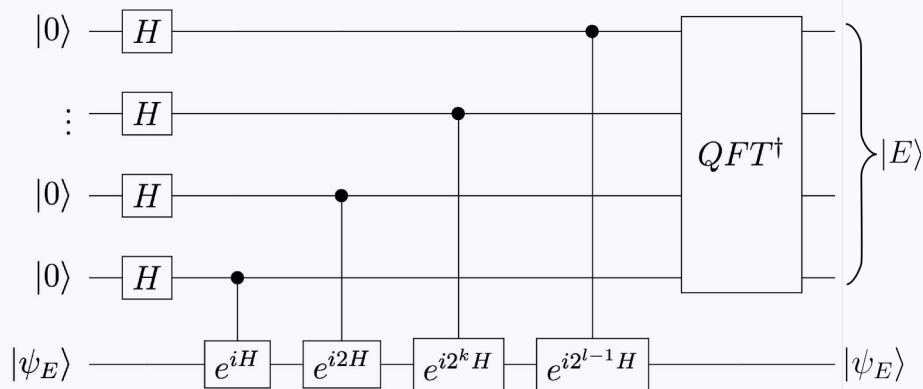
- Lower bound on total gate complexity^[2]

$\mathcal{O}(nt \text{ polylog}(nt/\epsilon)) \in \tilde{\Omega}(nt)$ number of gates for a n -qubit (time dep.) Hamiltonian

To go faster, exploit additional structure of the model!
symmetry, connectivity, spectral promise, ...

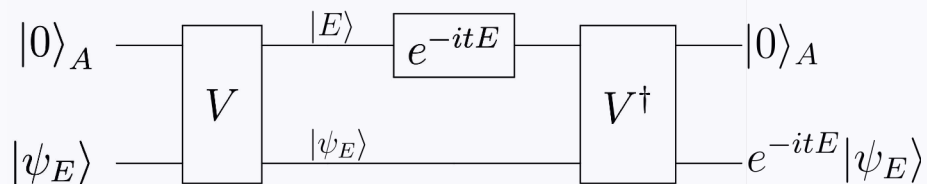
Spectral amplification for quantum simulations

1 Energy measurement (QPE)



↕ same gate complexity

2 Hamiltonian simulation circuit using V



→ Defines energy measurement circuit

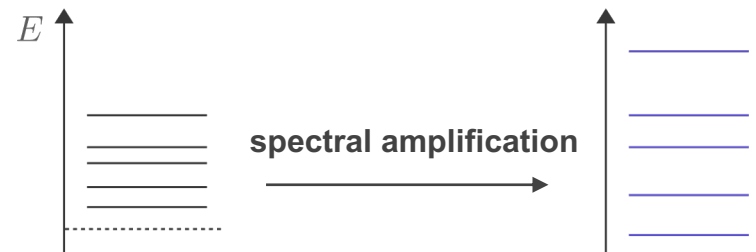
$$V |\psi_E\rangle |0\rangle_{\text{anc}} \approx |\psi_E\rangle |E_{\text{meas}}\rangle_{\text{anc}}$$

1 Longer time evolution for better precision

$$\delta E \sim 2^{-\ell} \sim 1/T$$

2 Precision of energy measurement determines simulation time for fixed error

$$t \sim 1/\delta E$$



**Better low-energy resolution
accelerates quantum simulation**

Quadratic speedup via squared form of the Hamiltonian

$$H = H_{\text{sqr}}^\dagger H_{\text{sqr}} + E_{\text{SA}} \implies \sigma_k(H_{\text{sqr}}) = \sqrt{E_k - E_{\text{SA}}}$$

■ Frustration-free Hamiltonian S. Gu, R. D. Somma, and B. Şahinoğlu, arXiv:2105.07304

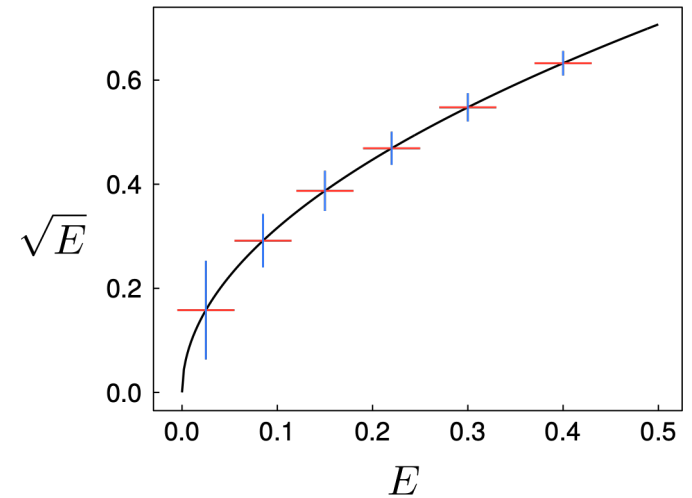
$$H = \sum_X h_X \quad (h_X \geq 0) \quad \rightarrow \quad H_{\text{sqr}} = \sum_X \sqrt{h_X} \otimes (|X\rangle\langle 0| + |0\rangle\langle X|)$$

■ Preprocessing by semidefinite programming (SDP)

$\mathbf{o}_a$: operator basis

$$\max_{G \geq 0, E_{\text{SA}}} E_{\text{SA}} \quad \text{subject to} \quad H - E_{\text{SA}} I = \sum_{a,b=1}^L G_{ab} \mathbf{o}_a^\dagger \mathbf{o}_b$$

$$H_{\text{sqr}} = \sum_m |m\rangle \otimes O_m \quad (H_{\text{sqr}}^\dagger H_{\text{sqr}} = \sum_m O_m^\dagger O_m)$$



Estimate of E to precision ϵ



Estimate of $\sqrt{E}$ to precision $\mathcal{O}(\epsilon/\sqrt{E})$

Related method for dissipative ODEs:

$$\dot{u} = -L^2 u, \quad u(T) = e^{-TL^2} u(0)$$

Query complexity with access to L : $\tilde{\mathcal{O}}(\alpha_L \sqrt{T})$

Query complexity with access to L^2 : $\mathcal{O}(\alpha_L T)$

D. An et al., arXiv:2211.05246

Application of spectral amplification to quantum chemistry

Molecular electronic Hamiltonian

$$H = \sum_{pq\sigma} h_{pq}^{(1)} a_{p\sigma}^\dagger a_{q\sigma} + \frac{1}{2} \sum_{pqmn, \sigma\tau} h_{pqmn}^{(2)} a_{p\sigma}^\dagger a_{q\sigma} a_{m\tau}^\dagger a_{n\tau}$$

Previous Result of
Low et al., Phys. Rev. X **15**, 041016 (2025)
(Google Quantum AI team)

Factorization by DFTHC

$$h_{pqmn}^{(2)} \approx \sum_{r \in [R], c \in [C]} \left(\sum_{b \in [B]} w_b^{(rc)} u_{bp}^{(r)} u_{bq}^{(r)} \right) \left(\sum_{b' \in [B]} w_{b'}^{(rc)} u_{b'm}^{(r)} u_{b'n}^{(r)} \right)$$

Tune (R, C, B) to reduce # fermionic rotations # QROAM table

■ Double Factorization (DF) $[(R, C, B) = (R, 1, N)]$

$$h_{pqmn}^{(2)} \approx \sum_r L_{pq}^{(r)} L_{mn}^{(r)} \quad \text{large rotation-data cost}$$

■ Tensor Hypercontraction (THC) $[(R, C, B) = (1, R_{\text{THC}}, R_{\text{THC}})]$

$$h_{pqmn}^{(2)} \approx \sum_{\mu\nu} X_{p\mu} X_{q\mu} Z_{\mu\nu} X_{m\nu} X_{n\nu} \quad \text{large coefficient lookup cost}$$

Further optimization with **BLISS**

Block-Invariant Symmetry Shift

$$H \rightarrow H + A(\hat{N} - \eta) + B(\hat{N} - \eta)^2$$

optimize A, B in the sector

$$\hat{N} |\psi_\eta\rangle = \eta |\psi_\eta\rangle$$

Same target-sector physics,
smaller block encoding normalization

Benchmark result for **FeMoco** [ground-state energy estimation at chemical accuracy]

FeMoco-76 [113e, 76o]

THC + BLISS (best prior)  4.3×10^9

DFTHC + BLISS + SA  1.00×10^9

4.3× fewer Toffolis

FeMoco-54 [54e, 54o]

Symmetry-compressed DF  2.6×10^9

DFTHC + BLISS + SA  3.42×10^8

7.0× fewer Toffolis

Limitations of SA: heavy classical preprocessing & data loading

Cost: classical preprocessing, # query $\times$ gate complexity of oracle

1 SDP preprocessing

heavy classical computation cost to find the square-root operator

e.g., $\mathcal{O}(N^{12})$, $\mathcal{O}(N^{18})$

for level-2, 3 SOS decomposition (using naïve optimization)



Direct SOS construction

2 Data loading

H_{sqr} may contain more terms than the original Hamiltonian

e.g., Hubbard model

$$H = -t \sum_{\langle i,j \rangle} \sum_{\sigma=\uparrow,\downarrow} (a_{i\sigma}^\dagger a_{j\sigma} + a_{j\sigma}^\dagger a_{i\sigma}) + U \sum_i a_{i\uparrow}^\dagger a_{i\uparrow} a_{i\downarrow}^\dagger a_{i\downarrow}$$

$\rightarrow \mathcal{O}(N)$ terms

$\mathcal{O}(N^4), \mathcal{O}(N^6)$ terms for level-2,3 SOS in the worst case



Generate data on demand

3 Normalization of H_{sqr}

Query complexity of quantum simulation using spectral amplification

figure of merit: $\lambda_{\text{eff}} = \sqrt{E_{\text{gap}} \lambda_{\text{sqr}}^2}$

$$E_{\text{gap}} = E_{\text{GS}} - E_{\text{SA}}$$

■ Better bound on GS energy



■ larger normalization by increasing # of coefficients



Efficient block encoding using Gaussian rotations

Solution

1. Supersymmetric fermion models
2. Direct SOS construction for random-coupling models

Direct application of spectral amplification to SUSY models

Fermionic SUSY models

fermion parity $P = (-1)^F$

supercharge $Q = Q^\dagger, \quad \{Q, P\} = 0$

Hamiltonian $H = Q^2$



$$H_{\text{sqr}} = Q, \quad E_{\text{SA}} = 0$$

- ✓ No SDP preprocessing
- ✓ Natural lower bound on GS energy

Example of SUSY models

$\mathcal{N}$: number of supercharges

■ Nicolai model ($\mathcal{N} = 2$) : $Q = \sum_j c_{2j-1} c_{2j}^\dagger c_{2j+1}, \quad H = \{Q, Q^\dagger\}$

■ SUSY SYK ($\mathcal{N} = 1$) : $Q = i \sum_{i < j < k} C_{ijk} \gamma_i \gamma_j \gamma_k, \quad H = Q^2$

■ SUSY SYK ($\mathcal{N} = 2$) : $Q = i \sum_{a < b < c} C_{abc} c_a c_b c_c, \quad H = \{Q, Q^\dagger\}$

For $\mathcal{N} \geq 2$,

$$Q = Q + Q^\dagger, \quad H = Q^2 \quad (\because Q^2 = 0)$$

$$\mathbb{E}[|C_{ijk}|^2] = \mathbb{E}[|C_{abc}|^2] = 2J/N^2$$

$\mathcal{N} = 2$ SUSY and topological data analysis (TDA)

SUSY	$Q^2 = 0$	—	$H = \{Q, Q^\dagger\}$	—	# of zero-energy states	$\ker H \cong \ker Q / \text{im } Q$
TDA	$\partial^2 = 0$	—	$\Delta = \{\partial, \partial^\dagger\}$	—	Betti numbers	$\beta_k = \dim \ker \Delta_k$

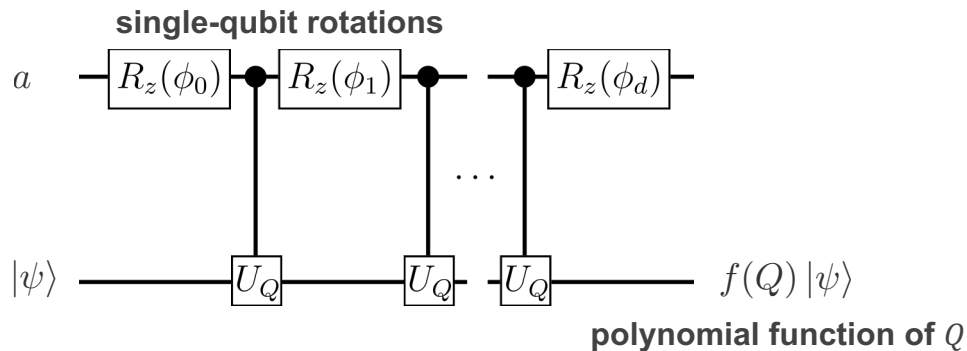
Hamiltonian simulation of SUSY models by QETU

Target: $H = Q^2$, $\{Q, P\} = 0$, $U = \exp(-iHt)$ ($0 \leq E \leq \Gamma$)

Oracle: time-evolution unitary by the supercharge $U_Q = \exp(-iQ/\lambda_{\text{sqr}})$

Quantum Eigenvalue Transformation of Unitaries (QETU)

constructing target function of Q by quantum signal processing (QSP)

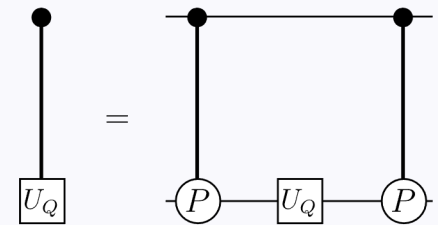


Interleave d calls to W with ancilla rotations

Approximate $f(Q) = \exp(-itQ^2)$ for $|Q| \leq \sqrt{\Gamma}$

- ✓ **One ancilla**
- ✓ **No block encoding**
- ✓ **No multi-qubit-controlled gates**

Forward-backward evolution



$$W = C_a(P)(I_a \otimes U_Q)C_a(P)$$

$$= |0\rangle\langle 0| \otimes U_Q + |1\rangle\langle 1| \otimes U_Q^\dagger$$

or

$$W = \exp(-iQ'/\lambda_{\text{sqr}})$$

$$Q' = \gamma_a \otimes Q$$

Q' : even operator

Total gate complexity reduction for SUSY SYK model

Polynomial approximation of $f(Q) = \exp(-itQ^2)$

$$Q|q\rangle = q|q\rangle, \quad |q| \leq \sqrt{\Gamma}, \quad \sqrt{\Gamma}/\lambda_{\text{sqr}} \leq \pi/12$$

Map the spectral interval to a fixed interval

QETU uses the variable $x = \sin(q/\lambda_{\text{sqr}})$.

Choose $\ell = \left\lfloor \frac{\pi\lambda_{\text{sqr}}}{12\sqrt{\Gamma}} \right\rfloor$ and apply the Chebyshev map:

$$y = Z_\ell(x) = \sin^2\left(\frac{2\ell q}{\lambda_{\text{sqr}}}\right) \in \left[0, \frac{1}{4}\right]$$

Approximate the phase in the new variable

$$e^{-itq^2} = \exp[-i\omega(\arcsin \sqrt{y})^2], \quad \omega = \frac{t\lambda_{\text{sqr}}^2}{4\ell^2} = \Theta(t\Gamma)$$

The degree of the polynomial $u(y)$

approximating this phase is $\deg u = \mathcal{O}\left(t\Gamma + \log \frac{1}{\epsilon}\right)$

Composing these polynomials,

$$\text{Query complexity: } d = 4\ell \deg u = \mathcal{O}\left(t\lambda_{\text{sqr}}\sqrt{\Gamma} + \frac{\lambda_{\text{sqr}}}{\sqrt{\Gamma}} \log \frac{1}{\epsilon}\right)$$

SUSY SYK benchmark

Trotterization to precision $\mathcal{O}(\epsilon/d) \times$ Query complexity d

$$G_{\text{1st}} = \tilde{\mathcal{O}}\left[\frac{N^6}{\epsilon} \left(t\sqrt{\Gamma} + \frac{1}{\sqrt{\Gamma}} \log \frac{1}{\epsilon}\right)^2\right], \quad G_{\text{high}} = N^{4+o(1)} \left(t\sqrt{\Gamma} + \frac{1}{\sqrt{\Gamma}} \log \frac{1}{\epsilon}\right)^{1+o(1)} \epsilon^{-o(1)}.$$

Standard quartic-SYK benchmark

$$N^{9/2}t$$

SA + QETU

$$N^4 t \sqrt{\Gamma}$$

Improvement (asymptotic scaling)

$$\sqrt{N/\Gamma}$$

Random coupling models allow efficient application of SA

Sum-of-squares construction of the SYK model

Blocks of quadratic Majorana operators

$$B_m = i \sum_{i < j} A_{ij}^{(m)} \gamma_i \gamma_j, \quad \{\gamma_i, \gamma_j\} = 2\delta_{ij}$$

Independent Gaussian entries

$$A_{ij}^{(m)} \sim \mathcal{N}(0, v), \quad v = \frac{J}{\sqrt{2MN^{3/2}}}$$

Family of random coupling models $\approx$ SYK model

$$H := \sum_{m=1}^M B_m^2 + E_{\text{SA}}, \quad E_{\text{SA}} = - \sum_m \sum_{i < j} (A_{ij}^{(m)})^2$$

➔

$$H = \sum_{i < j < k < l} J_{ijkl} \gamma_i \gamma_j \gamma_k \gamma_l, \quad J_{ijkl} = -2 \sum_m \left[A_{ij}^{(m)} A_{kl}^{(m)} + \text{anti-symmetric pairs} \right]$$
$$\mathbb{E}[J_{ijkl}] = 0, \quad \mathbb{E}[J_{ijkl}^2] = 6J^2/N^3$$

Low-rank SYK to full dense SYK model

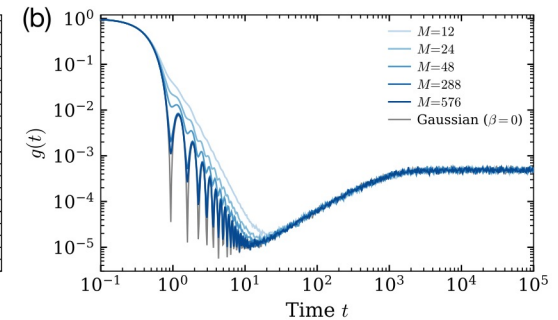
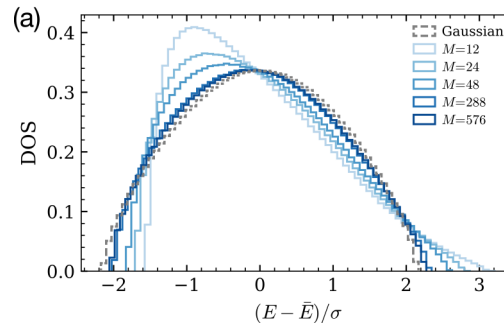
Kim, Cao, and Altman, Phys. Rev. B 101, 125112 (2020)

$$M = \Theta(N)$$

- Low-rank (class IV) SYK
- maximally chaotic
- tunable IR behavior (scaling dim.)

$$M = \Theta(N^\alpha) \quad (\alpha > 1)$$

- equivalent to dense SYK at large N (same SD equations)



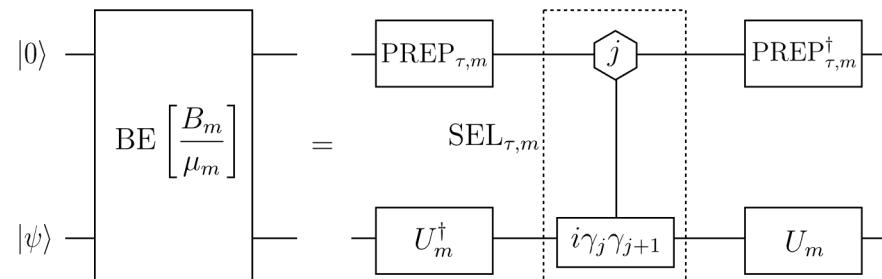
Block encoding of a single quadratic Majorana block

Reducing BE normalization cost by Gaussian rotation

$$B_m = U_m \left(\sum_{j=1}^{N-1} \tau_j^{(m)} i\gamma_j \gamma_{j+1} \right) U_m^\dagger$$

random Gaussian rotation

$$\frac{|\tau_j^{(m)}|}{\sqrt{v}} \sim \chi_{N-j}; \quad \chi_k : \text{length of } k \text{ Gaussian variables}$$



Direct linear combination of unitaries (LCU)

$\Theta(N^2)$ coefficients

$$\sum_{i < j} |A_{ij}^{(m)}| = \Theta(N^2 \sqrt{v})$$

$\sqrt{N}$ improvement

Tridiagonal representation

$N - 1$ coefficients

$$\mu_m = \sum_j |\tau_j^{(m)}| = \Theta(N^{3/2} \sqrt{v})$$

Prepare and Select oracle

$$\text{PREP}_{\tau,m} |0\rangle = \frac{1}{\sqrt{\mu_m}} \sum_{j=1}^{N-1} \sqrt{|\tau_j^{(m)}|} |j\rangle$$

$$\text{SEL}_{\tau,m} \sum_{j=1}^{N-1} |j\rangle\langle j| \otimes i\gamma_j \gamma_{j+1}$$

Random coefficient generator

$$O_\tau |m, j\rangle |0\rangle = |m, j\rangle |\tau_j^{(m)}\rangle$$

$$O_\theta |m, k\rangle |0\rangle = |m, k\rangle |\theta_k^{(m)}\rangle$$

angles of the elementary rotations for random Gaussian unitaries

(approximate) random numbers generated in $\text{polylog}(N, M)$ gate complexity

Block encoding of the square root operator $H_{\text{sqr}} = \sum_m |m\rangle \otimes B_m$

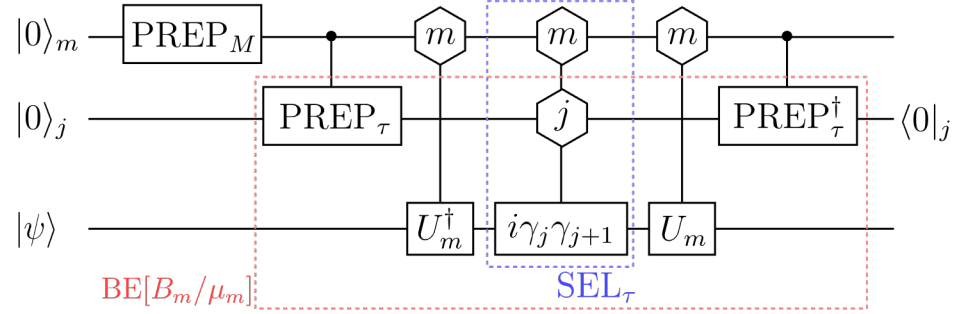
$$\lambda_{\text{sqr}}^2 = \sum_{m=1}^M \mu_m^2$$

$$\text{SEL}_\tau = \sum_{m,j} |m, j\rangle \langle m, j| \otimes i\gamma_j \gamma_{j+1}$$

$$\text{PREP}_M |0\rangle_m = \frac{1}{\lambda_{\text{sqr}}} \sum_{m=1}^M \mu_m |m\rangle_m$$

$$\text{PREP}_\tau |m\rangle_m |0\rangle_j = |m\rangle_m \frac{1}{\sqrt{\mu_m}} \sum_{j=1}^{N-1} \sqrt{|\tau_j^{(m)}|} |j\rangle_j$$

$$\mathcal{U} = \sum_{m=1}^M |m\rangle \langle m| \otimes U_m$$



$$\mathcal{V} := \text{PREP}_\tau^\dagger \mathcal{U} \text{SEL}_\tau \mathcal{U}^\dagger \text{PREP}_\tau \text{PREP}_M$$

$$(I_m \otimes \langle 0|_j \otimes I) \mathcal{V} (|0\rangle_m \otimes |0\rangle_j \otimes I) = \frac{H_{\text{sqr}}}{\lambda_{\text{sqr}}}$$

all constructed from the elementary oracle O_τ, O_θ

On-the-fly implementation of random Gaussian unitaries

$$U_m = G_K(\theta_K^{(m)}) \cdots G_2(\theta_2^{(m)}) G_1(\theta_1^{(m)}), \quad K = \frac{N(N-1)}{2}$$

$$|m, k, 0\rangle |\psi\rangle \xrightarrow{O_\theta} |m, k, \theta_k^{(m)}\rangle |\psi\rangle \xrightarrow{G_k(\theta_k^{(m)})} |m, k, \theta_k^{(m)}\rangle G_k(\theta_k^{(m)}) |\psi\rangle \xrightarrow{O_\theta^\dagger} |m, k, 0\rangle G_k(\theta_k^{(m)}) |\psi\rangle.$$

Total gate complexity for the SYK model

Input energy window $E \in [0, \Gamma]$ ($\Gamma \sim -E_{\text{SA}}$)

GS energy lower bound $-E_{\text{SA}} = \Theta(MN^2v) = \Theta(J\sqrt{MN})$

BE normalization $\lambda_{\text{sqr}}^2 = \Theta(MN^3v) = \Theta(J\sqrt{MN^3})$

Overall Gate complexity

$$d = \underbrace{\mathcal{O}\left(t\sqrt{\Gamma\lambda_{\text{sqr}}^2} + \sqrt{\lambda_{\text{sqr}}^2\Gamma^{-1}}\log\frac{1}{\epsilon}\right)}_{\text{Query complexity}} = \mathcal{O}\left(JtN\sqrt{M} + \sqrt{N}\log\frac{1}{\epsilon}\right) \times \underbrace{\tilde{\mathcal{O}}(N^2)}_{\text{Gate complexity of oracle}}$$



$$G = \tilde{\mathcal{O}}\left(JtN^3\sqrt{M} + N^{5/2}\log\frac{1}{\epsilon}\right)$$

$\sqrt{N}$ improvement for fully dense SYK regime

Standard quartic-SYK benchmark
(higher-order Trotterization)

$$N^{9/2+o(1)}t^{1+o(1)}\epsilon^{-o(1)}$$

Our work at $M = \Theta(N^2)$

$$\tilde{\mathcal{O}}(N^4t)$$

Improvement (asymptotic scaling)

$$\sqrt{N}$$

- ✓ **No SDP**
- ✓ **Reduced BE normalization**
- ✓ **Lower oracle gate complexity**

Summary and Outlook

Solution

1. Supersymmetric fermion models
2. Direct SOS construction for random-coupling models

1 SDP preprocessing

heavy classical computation cost to find the square-root operator

Direct SOS construction

SUSY models

$$H = Q^2, \{Q, P\} = 0$$

sum of random blocks

$$H = \sum_{m=1}^M B_m^2 + E_{\text{SA}}$$

2 Data loading

H_{sqr} may contain more terms than the original Hamiltonian

Generate data on demand

pseudo random coefficient generators

$$O_\tau |m, j\rangle |0\rangle = |m, j\rangle |\tau_j^{(m)}\rangle$$

$$O_\theta |m, k\rangle |0\rangle = |m, k\rangle |\theta_k^{(m)}\rangle$$

3 Normalization of H_{sqr}

Query complexity of quantum simulation using spectral amplification

$$\text{figure of merit: } \lambda_{\text{eff}} = \sqrt{E_{\text{gap}} \lambda_{\text{sqr}}^2}$$

$$E_{\text{gap}} = E_{\text{GS}} - E_{\text{SA}}$$

Efficient block encoding using Gaussian rotations

$$B_m = U_m \left(\sum_{j=1}^{N-1} \tau_j^{(m)} i\gamma_j \gamma_{j+1} \right) U_m^\dagger$$

Summary and Outlook

SUSY quantum mechanics with dynamical bosons

$$W = \sum_{ijk} C_{ijk} x_i x_j x_k, \quad \mathcal{Q} = \frac{1}{\sqrt{2}} \sum_i c_i^\dagger (p_i - i\partial_i W)$$

$$H = \{\mathcal{Q}, \mathcal{Q}^\dagger\} = \mathcal{Q}^2$$

Open quantum system with Hermitian jump operators

$$\mathcal{L}(\rho) = -\frac{1}{2} \sum_m [B_m, [B_m, \rho]], \quad C_m = B_m \otimes I - I \otimes B_m^\top,$$

$$-\hat{\mathcal{L}} = \frac{1}{2} \sum_m C_m^\dagger C_m \quad \text{vectorization}$$

Diffusion on a random network

$$\partial_t u = -Lu, \quad L = D^\top K_e D = B^\dagger B \quad \text{random conductance on each edge}$$

Stochastic differential equations

$$dX_t = -KX_t dt + \sigma dW_t \quad K = \sum_e B_e^\dagger B_e$$